SUSHANTH YELISHETTY

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Education

Indian Institute of Technology (IIT), Patna *Expected May 2028*
B. Tech in Mechanical Engineering and M. Tech in Mechatronics (Dual Degree)

Skills

Languages: C, C++, Python, MATLAB, JavaScript, Bash, HTML, CSS, Android Java, SQL
Frameworks: ROS2, Gazebo, Rviz2, Habitat-Sim, C++ STL, TensorFlow, Keras, PyTorch, Scikit-learn, Stable-Baselines3, LightGBM, SentenceTransformers, PyMuPDF, NLTK, SpaCy, OpenCV, NumPy, Pandas, Matplotlib, Seaborn, Tkinter, Edge Impulse (TinyML)
Miscellaneous: Docker, ROS, Microcontrollers, Linux/UNIX Systems, Git, MySQL

Experience

Summer Research Intern May 2026 – Jul 2026
Robotics Research Centre (RRC), IIIT Hyderabad *Hyderabad, India*
- Built an end-to-end ROS 2 navigation pipeline (SLAM + Nav2) for Habitat-Sim and Gazebo, enabling autonomous mapping and goal-directed navigation in indoor environments.
- Engineered a custom CycloneDDS bridge across a Python 3.9/3.10 boundary to synthesize LIDAR, odometry, and TF streams between Habitat-Sim and the ROS 2 stack.
- Implemented a Model Predictive Control (MPC) local planner that optimizes control trajectories over a receding horizon for smooth obstacle avoidance around static and dynamic obstacles and goal-reaching within the Nav2 stack.

Robotics Software Support Engineer (Intern) Dec 2024 – Mar 2025
Magnify Biosciences Inc. *Pittsburgh, PA (Remote)*
- Handled programming, device support, deployment, and debugging for Raspberry Pi 4 and 5, utilizing IoT-based programs and libraries for device-specific functionality.
- Completed the device software and achieved successful end-to-end deployment of the device.

Projects

VLM-Driven Object-Goal Navigation
- Built an open-vocabulary ROS 2 navigation system where a mobile robot localizes and drives to objects named in natural language, using a local vision-language model (YOLO-World) with HSV refinement on the live RGB-D stream.
- Implemented camera-to-map back-projection (pinhole intrinsics + tf2) and a search-then-confirm behavior loop integrating Nav2, slam toolbox, and AMCL across ROS2 nodes in Gazebo.

Robotics Simulation and Navigation Pipeline (ROS2 Humble, Habitat-Sim, Nav2, CycloneDDS)
- Developed a custom CycloneDDS bridge linking Habitat-Sim to ROS2 across separate Python runtimes, enabling camera-only navigation validated against a lidar-SLAM baseline map.
- Deployed a full Nav2 stack (slam toolbox mapping, AMCL localization, costmap-based planning) for autonomous object-goal navigation in reconstructed indoor scenes.

Model Predictive Control (MPC) Local Planner
- Built a nonlinear MPC planner in Python (CasADi + IPOPT) for a unicycle-model robot, optimizing speed and turn-rate over a receding horizon while respecting actuator limits.
- Added static and dynamic obstacle avoidance using signed-distance constraints, propagating moving obstacles' predicted motion across the horizon for collision-free navigation.

Real-Time Facial Emotion Recognition
- Built a CNN-based facial emotion recognition system in PyTorch, trained on FER2013 with data augmentation and live webcam inference using OpenCV and multithreaded processing.
- Deployed an interactive Gradio interface for real-time inference and evaluation across diverse expressions and lighting conditions.

Embedded OCR on Edge Devices
- Developed an OCR pipeline using PyTorch CRNN with CTC loss for constrained embedded hardware.
- Applied TensorRT quantization and deployed on ESP32, integrating adaptive thresholding and character segmentation.

Deep Learning-Based AI Artifact Detection
- Developed a PyTorch ResNet-50 model with OpenCV preprocessing to detect AI-generated images via high-frequency artifact analysis.
- Implemented DCT coefficient engineering and Grad-CAM interpretability, achieving 92% AUROC against Stable Diffusion and DALL·E outputs.